Hardware Security

硬體安全

Introduction

Wei-Chih Hong

Information Security

- Wikipedia
  - **Information security**, sometimes shortened to InfoSec, is the practice of defending information from unauthorized access, use, disclosure, disruption, modification, perusal, inspection, recording or destruction. It is a general term that can be used regardless of the form the data may take (e.g. electronic, physical).

Computer Security

- **The NIST** *Computer Security Handbook*
  - The protection afforded to an automated information system in order to attain the applicable objectives of preserving the integrity, availability, and confidentiality of information system resources (includes hardware, software, firmware, information/data, and telecommunications).

This course will be focusing on Hardware Security

Hardware Security?

Valuable Information inside Hardware

- Smartphone
- Smart card
- Set-top box
- Game console

Related Topics

- Cryptography
- Hardware designs (VLSI, FPGA, embedded systems)
- Physical attacks and tamper resistance
- Hardware metering
- Physical Unclonable Functions (PUF)
- Hardware trojan

Attack Scenario

- Theft of service, private data, even money
- Cloning and overbuilding
- IP privacy
- Denial of service

How to design a secure system?

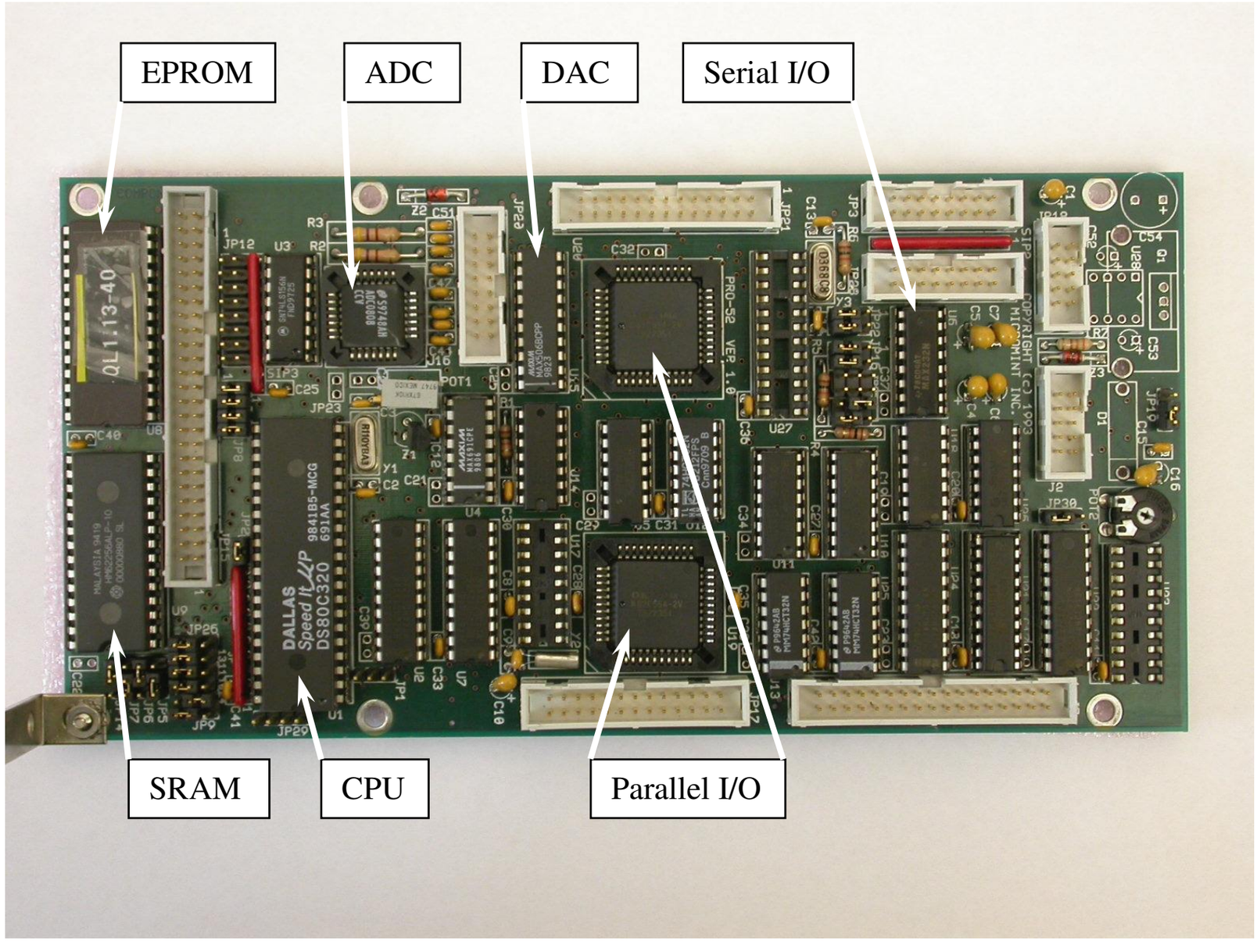
- What are the reasons to attack your system?
- Who is likely to attack your system?
- What tool would they use for the attacks?
- How to protect against these attacks?

- “A determined hacker can break any protection provided he has enough time and resources. The question is how practical it would be.”
- “It is impossible to avoid all problems; a reasonable goal is to make the process of breaking your design more expensive and time-consuming. With luck, potential attackers will switch to other products rather than spending money and effort on breaking yours.”

- “You have to find a trade off between the price you will pay for the extra security and the actual protection it will give you. In the end you have to think about who will be likely to attack your device.”
- “The only practical and reliable way of comparing security in different microcontrollers and smartcards is by estimating the cost of a successful attack which will involve actual breaking into the devices.”

Security Evolution in Silicon Chips

- Thirty years ago, no protection at all



Security Evolution in Silicon Chips

- Much smaller ASICs or MCUs, but still with external memories

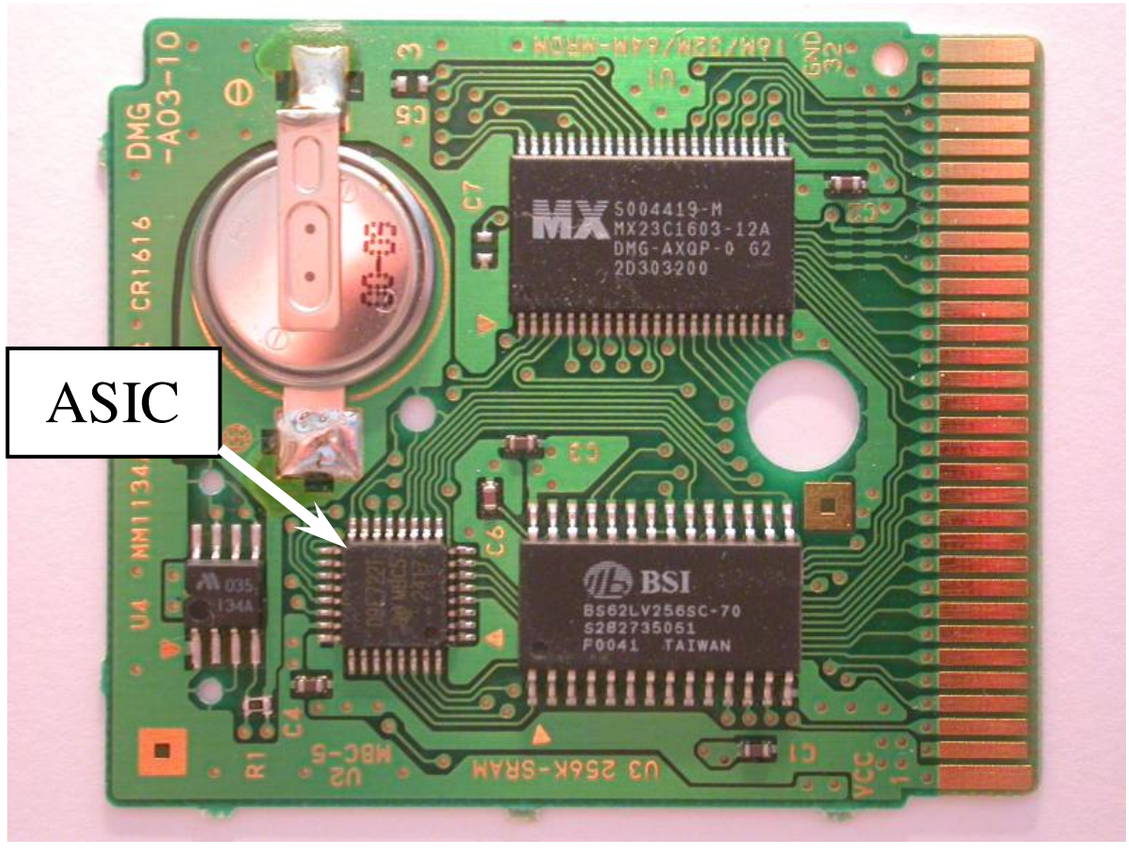


Figure 2. Game cartridge for Nintendo GameBoy game console [20]

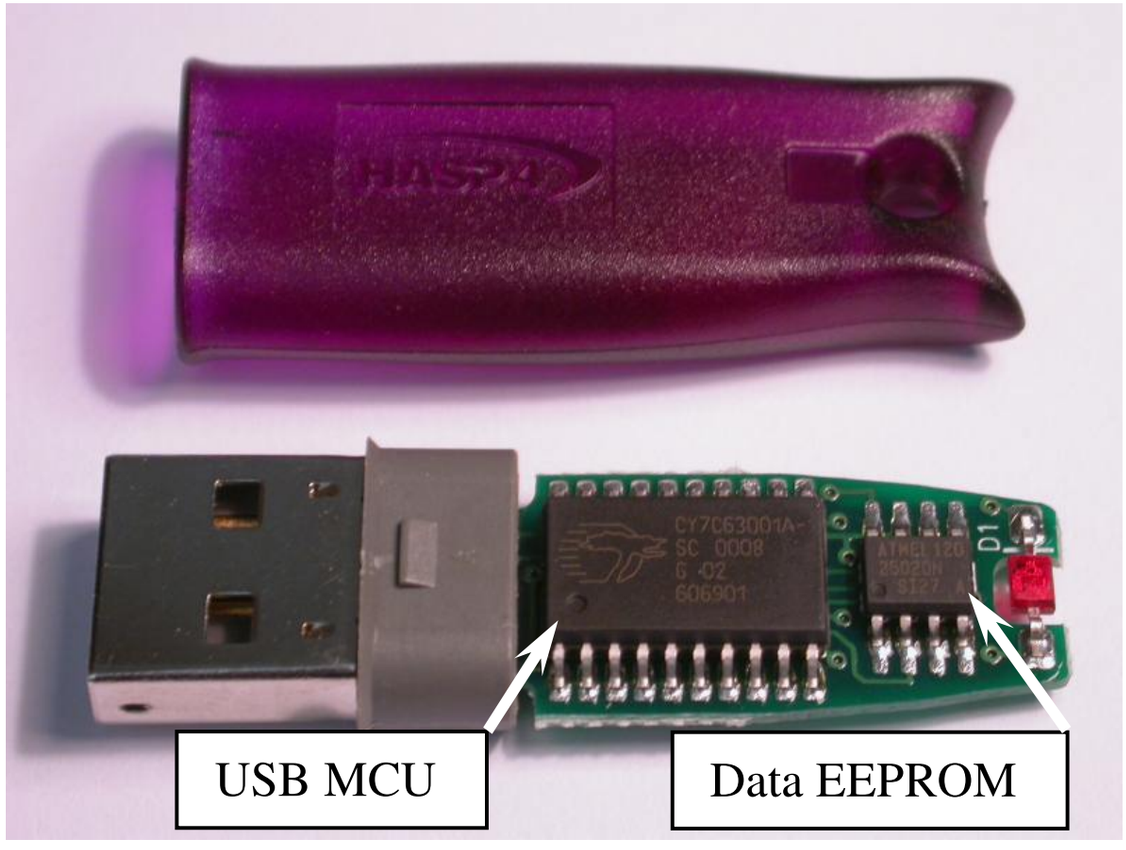


Figure 3. Aladdin HASP4 USB dongle [21]

Security Evolution in Silicon Chips

- MCUs with internal memories

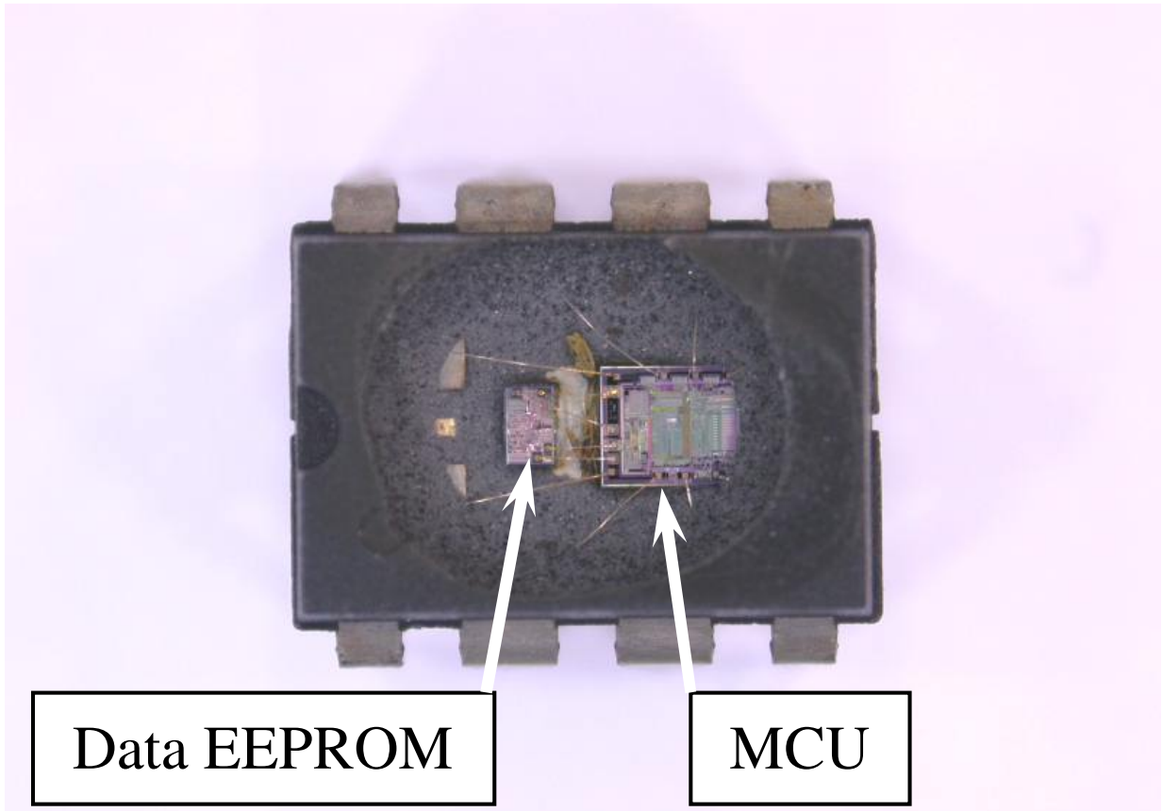


Figure 4. Non-volatile data memory and micro-controller chip are encapsulated into the same package in Microchip PIC12CE518 microcontroller [22]

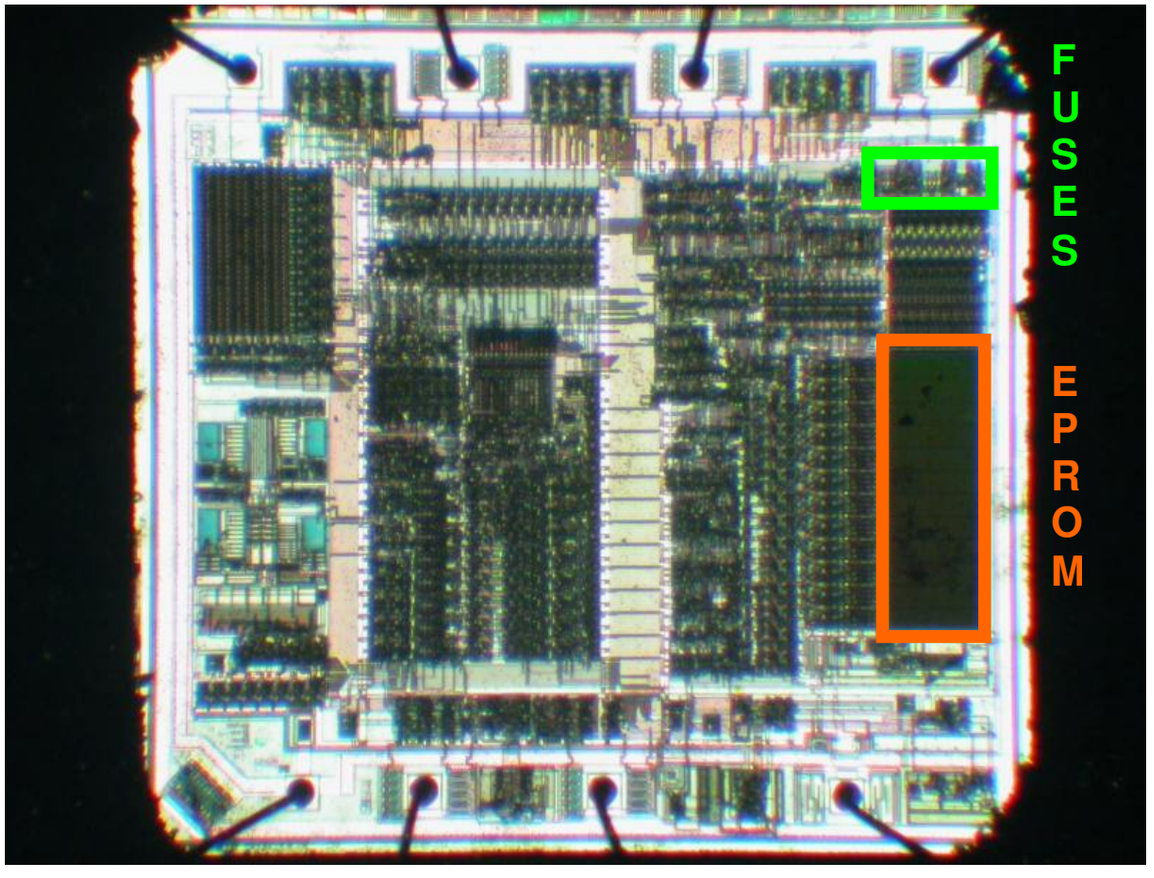
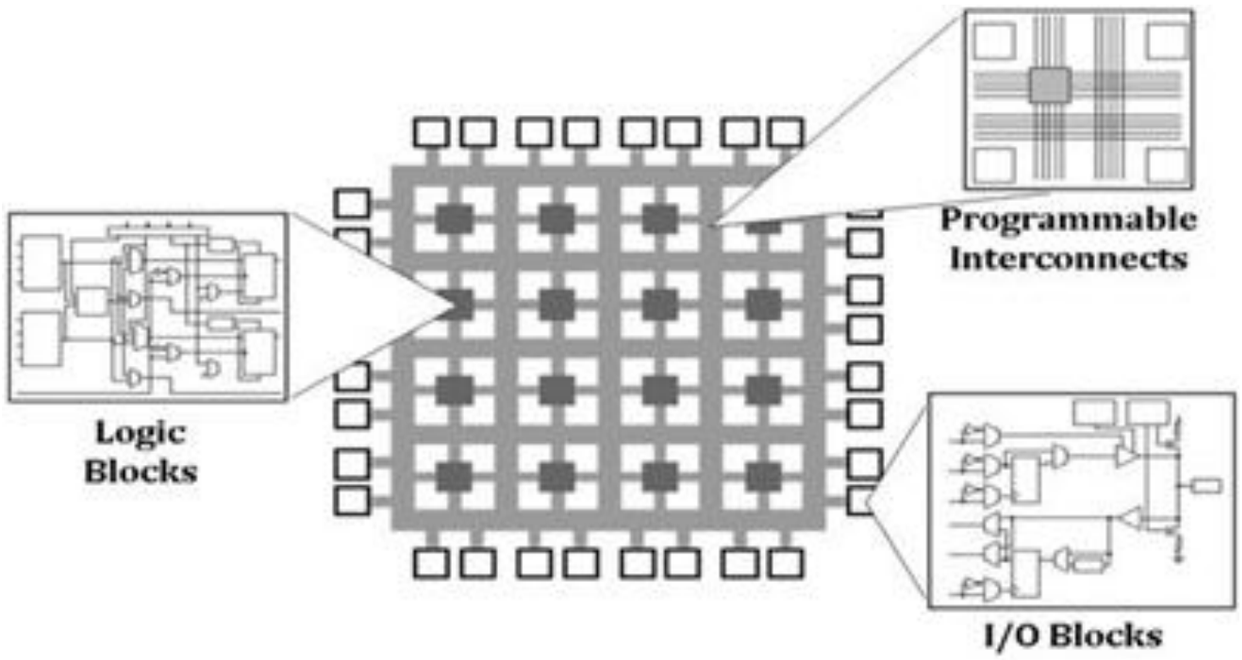


Figure 5. Security fuses are located outside the program memory array in Microchip PIC12C508 microcontroller [22] and can be easily disabled with UV light

Categories of Implementations

- Micro-controller
- FPGA (Field Programmable Gate Array)
- ASIC (Application Specific Integrated Circuit)

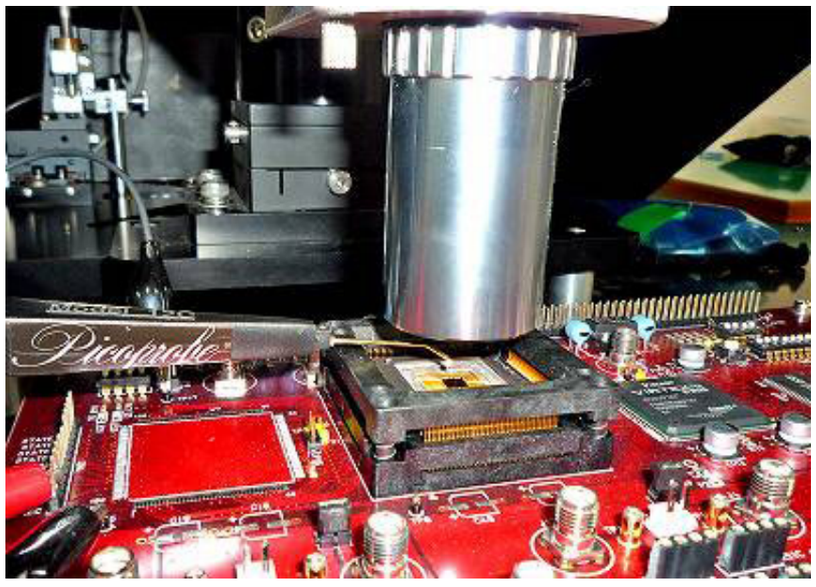


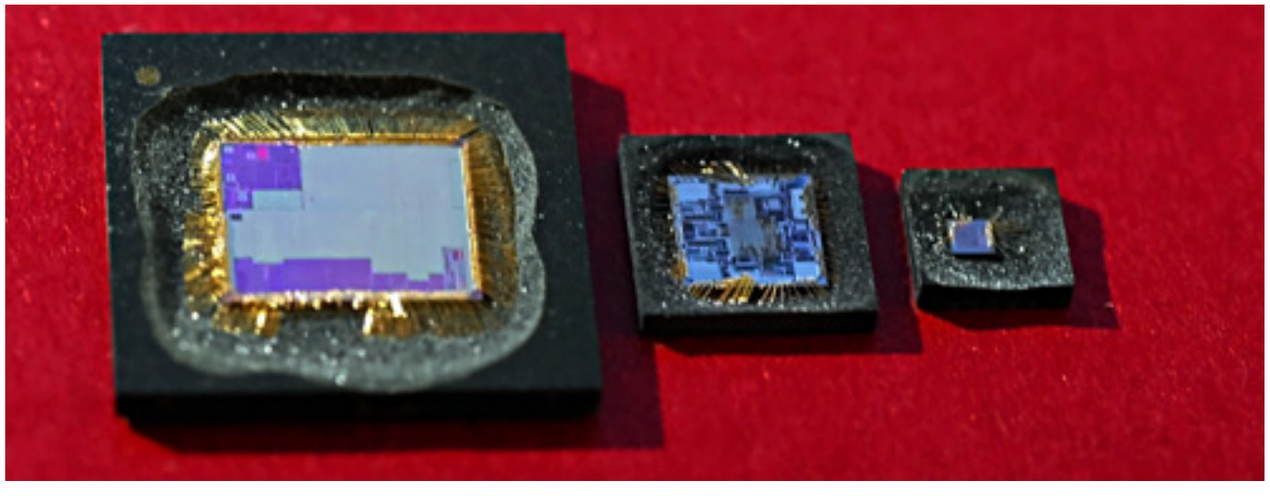
Attack Categories

- Software attack
- Side channel attack
- Fault injection
- Micro-probing
- Reverse engineering

Attack Categories

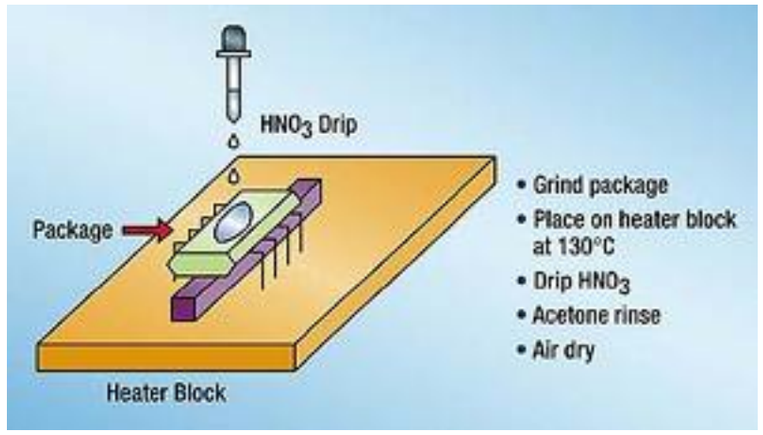
- Passive vs. active
- Interfaces of attacks
  - Invasive
  - Non-invasive
  - Semi-invasive



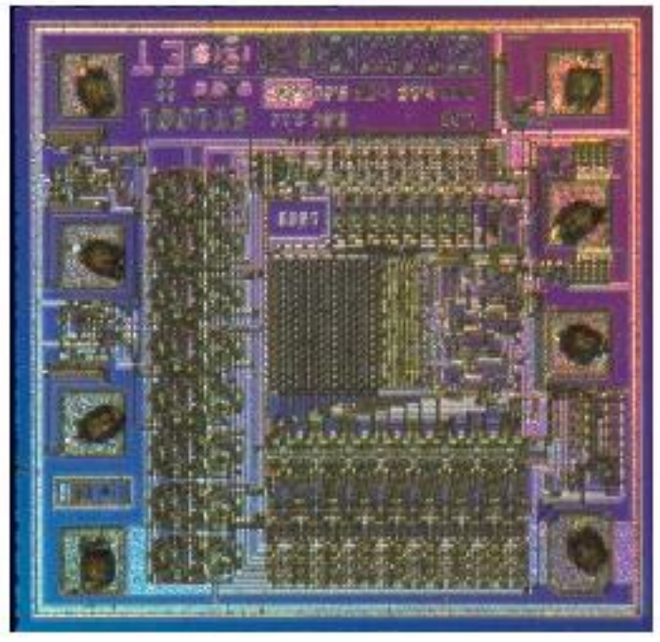


Invasive Attack

- Package decapsulation

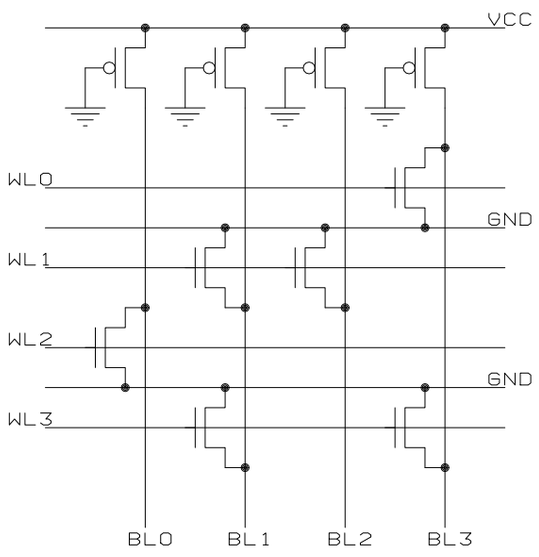


- Optical reading
- Micro-probing
- Focused Ion Beam (FIB)



Optical Reading the Mask ROM

Active layer programming



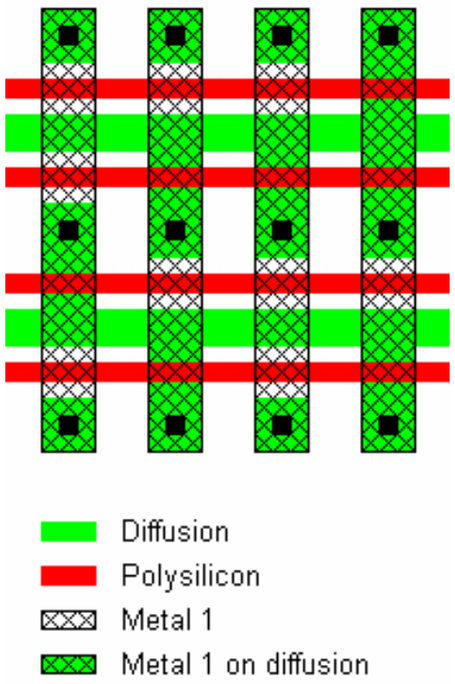
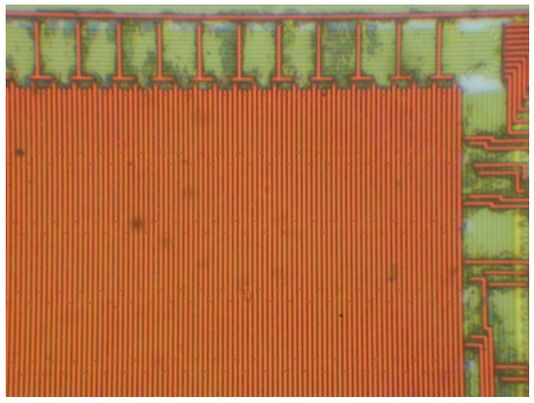
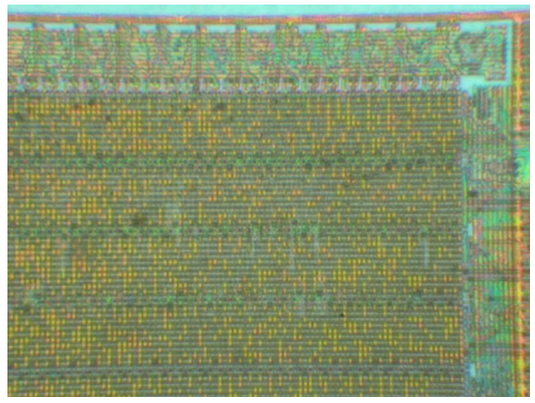


Figure 18. Configuration and layout of MOS NOR ROM with active layer programming. This type of memory can be read optically





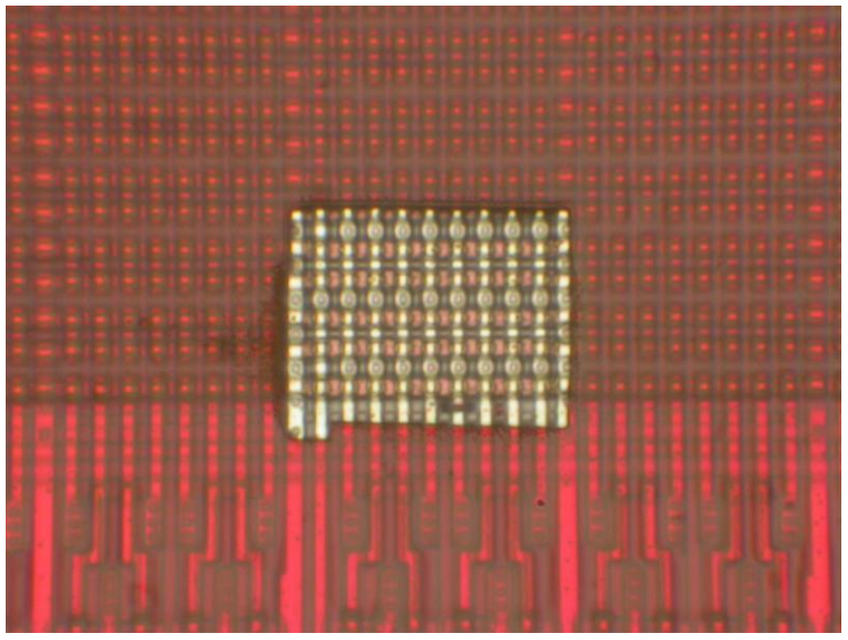
Removing top level metal for direct observation of data in NEC MCU

Semi-invasive Attack

- Introduced by Sergei Skorobogatov
- Require decapsulation, but without creating contacts to internal lines
- **UV light attack**
  - Optical fault injection

Semi-invasive Attacks

- Toothpick Attack
  - Low cost, easy to learn
  - Can be used to find the security fuses
    - 10um precision at most
  - Works on EPROM chips

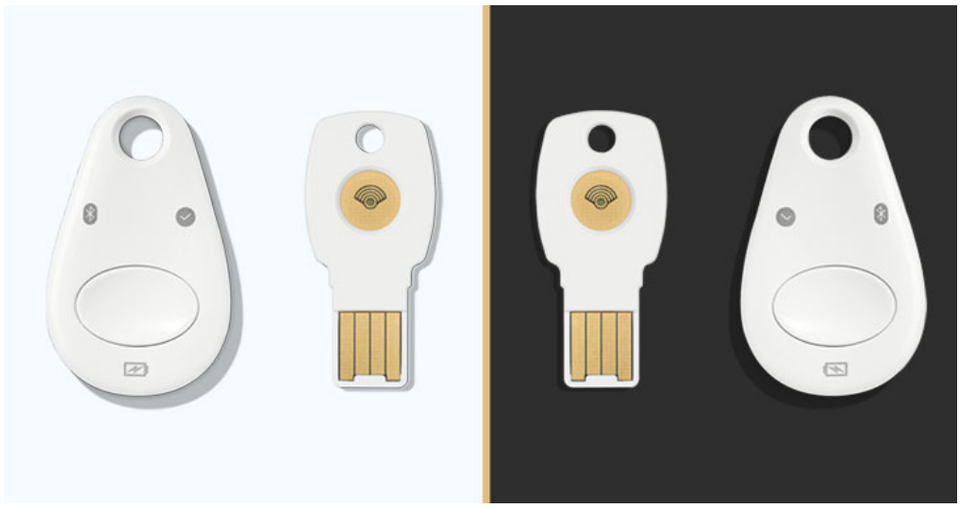


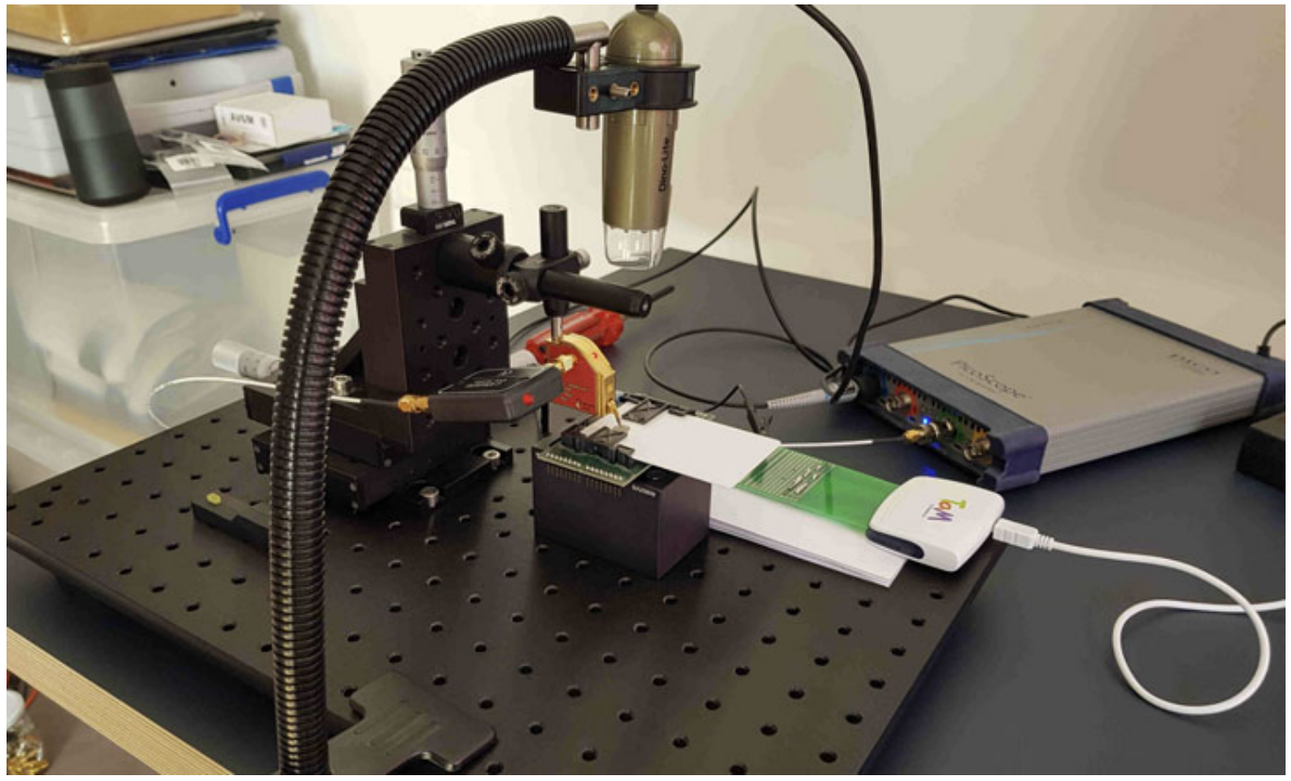
Non-invasive Attack

- **Most serious threat to hardware security**
  - Relatively low cost (usually)
  - No tamper evidence
- **Passive**
  - Side channel attack
- **Active**
  - Fault injection attack (glitch attack)
  - Data remanence

A Side Journey to Titan

- **Hardware security experts broke 2FA devices like Google Titan Key or Yubikey with side channel analysis techniques.**
  - ECDSA based on NXP chips





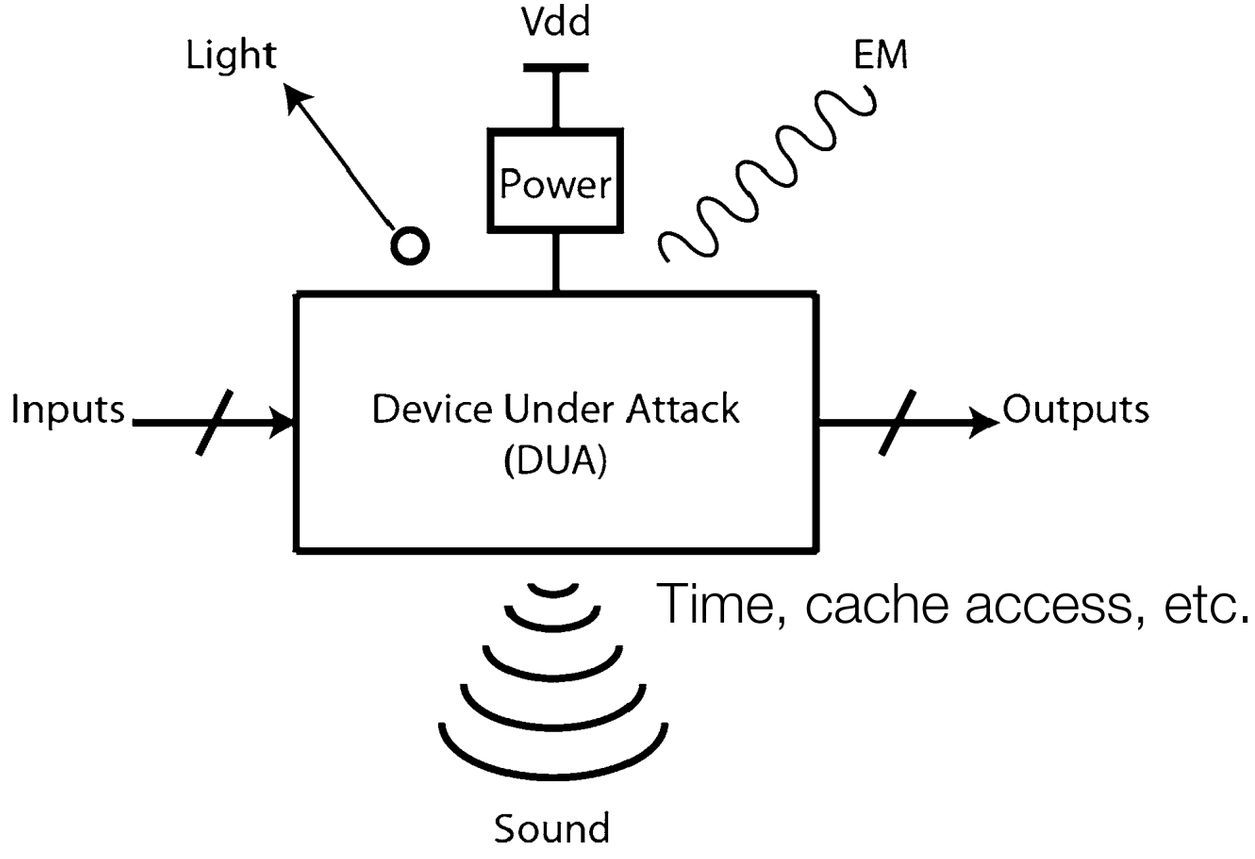
https://thehackernews.com/2021/01/new-attack-could-let-hackers-clone-your.html?&web_view=true

Introduction to Side Channel Attacks

Your system is as secure as its weakest link.

Side Channel Attack

- Bypass the theoretical strength of cryptographic algorithms
- Side channel information

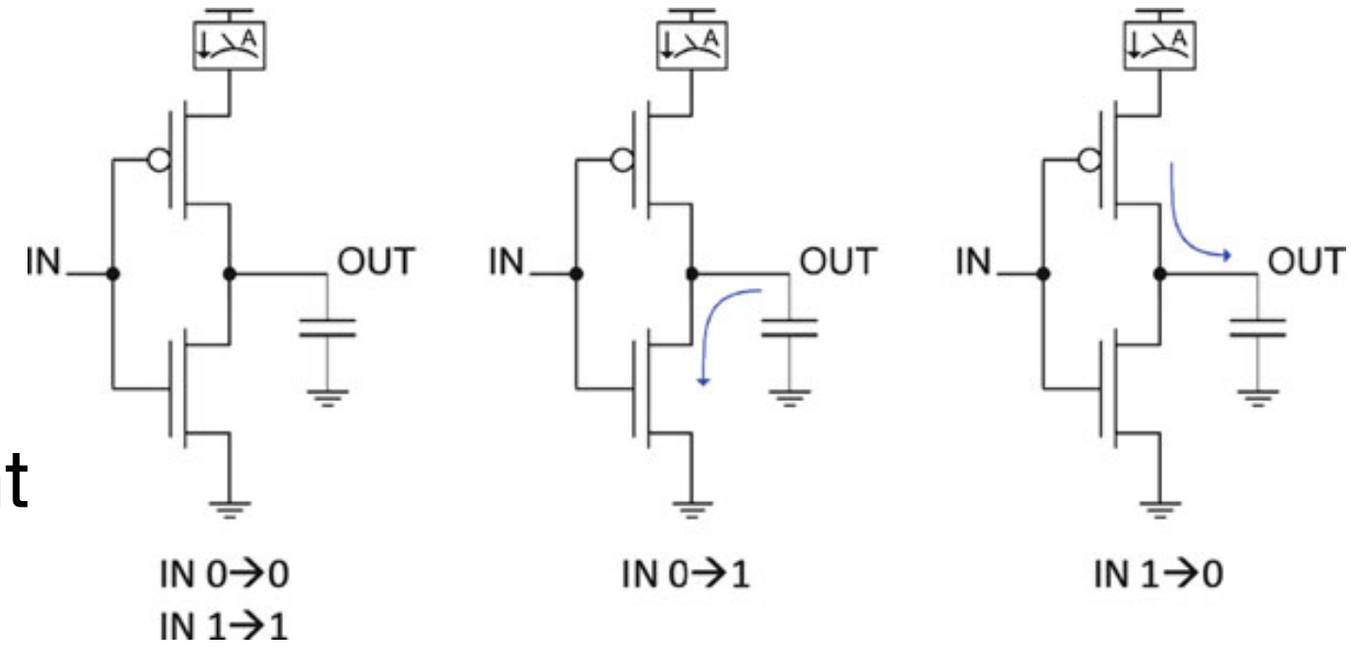


Brief History

- Researches started during World War II
  - Bell Labs, 1943
- NSA TEMPEST program
- Academic and industrial researches started from the 1990’s

Power Consumption

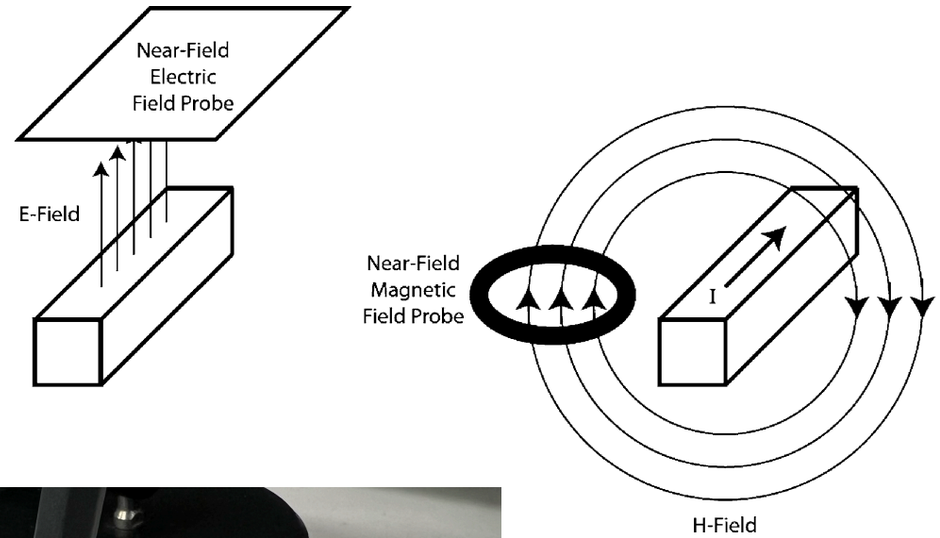
- **CMOS inverter**
  - Dynamic power
    - charging current
    - short circuit current
    - glitches
  - Static power
- **Measurement point**
  - Voltage regulator module on PCB

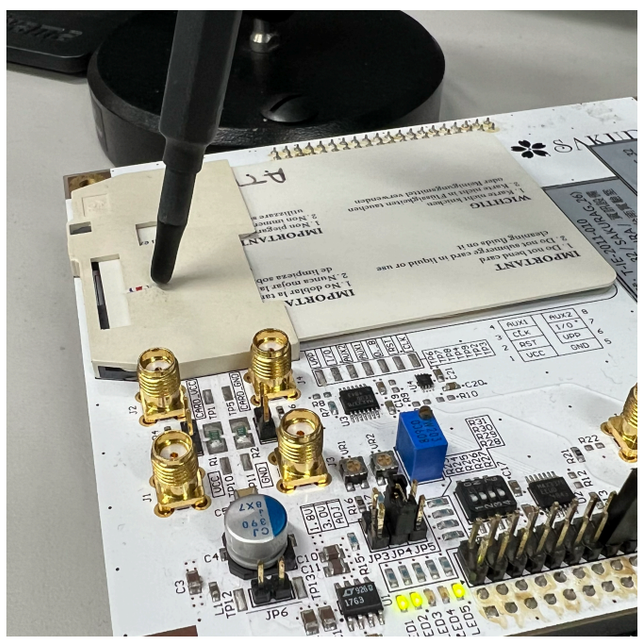


$$P_{dynamic} = \frac{1}{2} C V_{dd} \Delta V f$$

Electro-Magnetic Emission

- **Near-field: within two wavelengths**
  - 1MHz ⇒ 600m; 10GHz ⇒ 6cm
  - EM field induction
- **Far-field**
  - Radiated EM wave
- **Modulated signals**
  - AM, FM



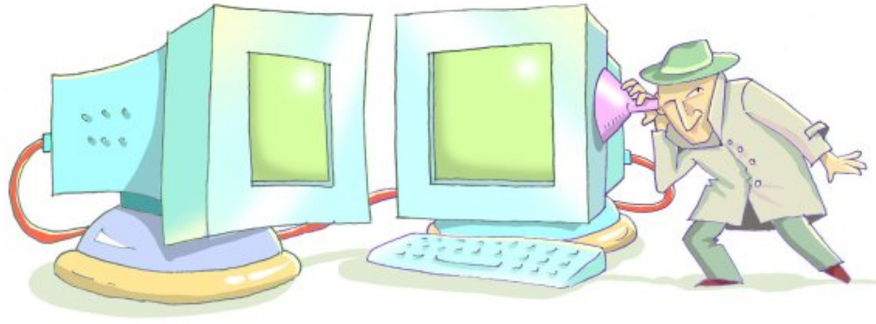


EM Attack

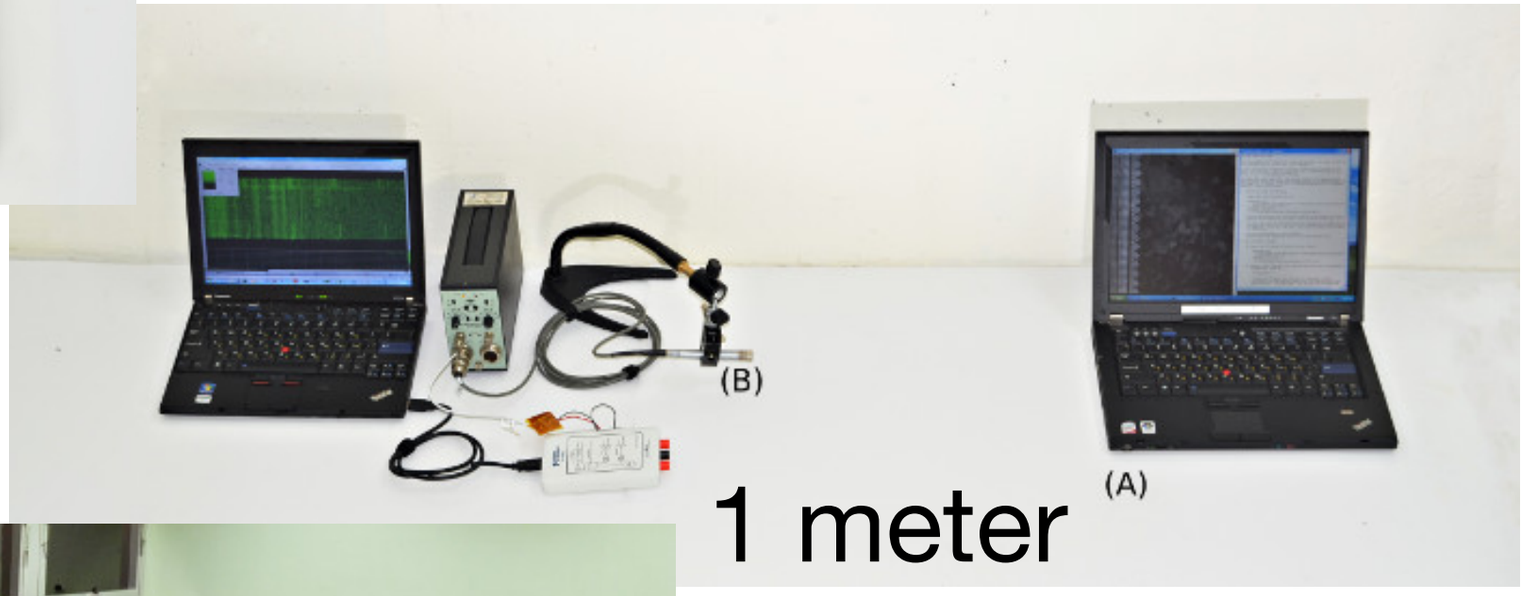
- **Measurement**
  - Carefully placed and oriented sensor(s)
- **Attacks**
  - Simple EMA
  - Differential EMA
- **Relationship with power analysis**
  - Power analysis countermeasures do not necessarily reduce EM emission!

Acoustic Side Channel

- Key wheels of Egyptian encryption machine
- Text printed by dot matrix printers
- Keyboard
- **RSA running on a PC**
  - Detect RSA execution and distinguish different keys
  - Piezoelectric properties of ceramic capacitors on the motherboard



Attack implemented by Shamir and Tromer



1 meter

4 meters

https://cs-people.bu.edu/tromer/acoustic/

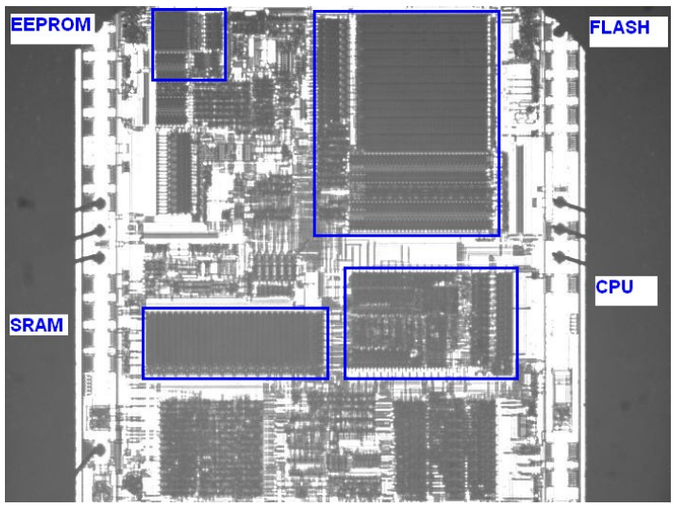
Optical Emission

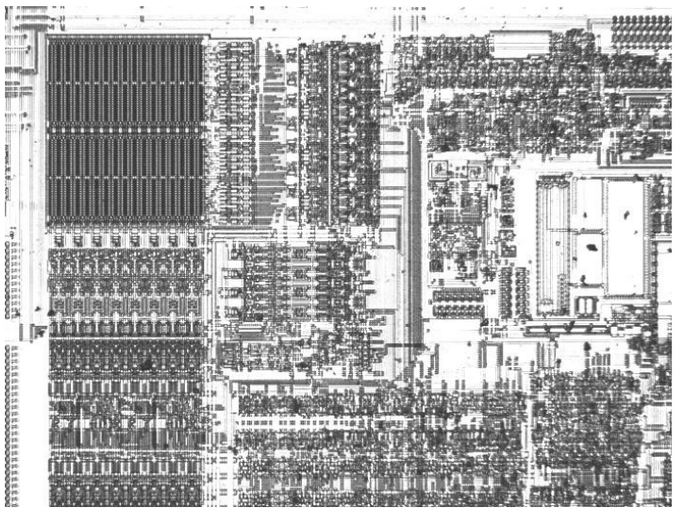
- **Visible or infrared light emission**
- **Low emission, low prob. of capture**
  - need for repeated cycles with the same inputs
- **Measurement**
  - photomultipliers or CCD camera
  - laser voltage probing (LVP) system

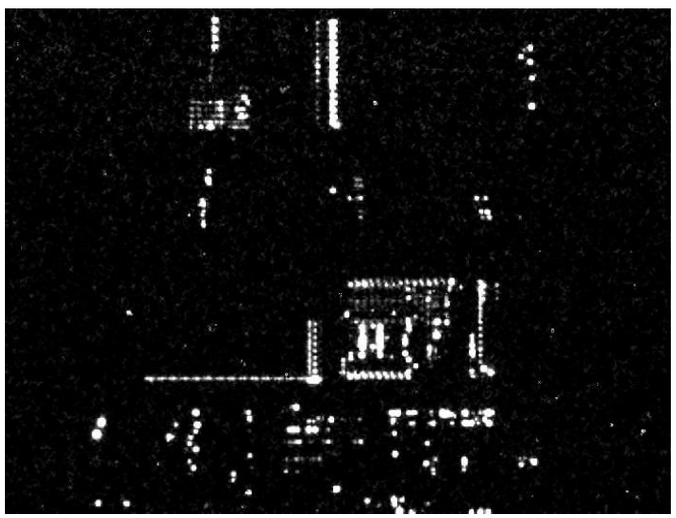
Optical Side-Channel

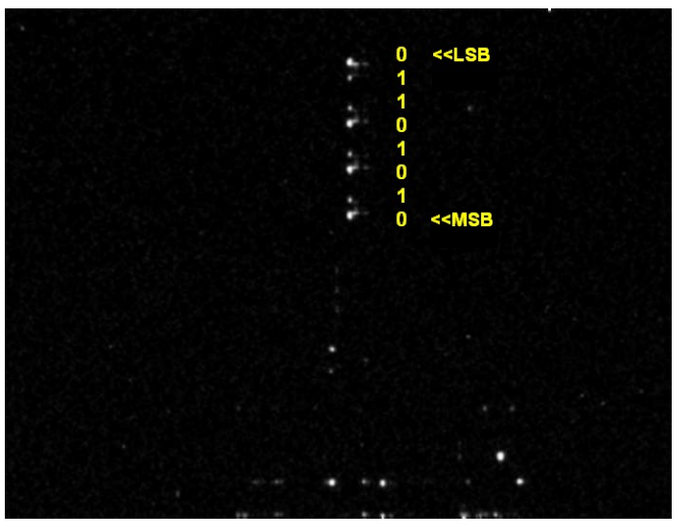
- Transistors emit photons when they switch
- Optical emission can be detected with photomultipliers and CCD cameras

Microchip PIC16F628 MCU with test code at 20 Mhz



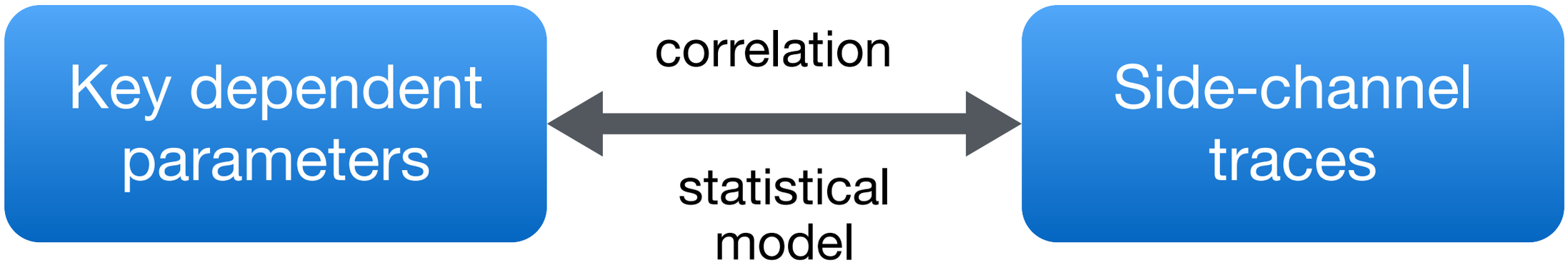






S. Skorobogatov's work at FDTC 2009

Strategies of SCA



- Assume a hypothetical model that relates internal values or operations to the side-channel signals.
- Differential analysis combats the noises with the strength of statistical techniques.
- Profiled attacks “learn” the parameters of the model based on the traces recorded from the golden sample.

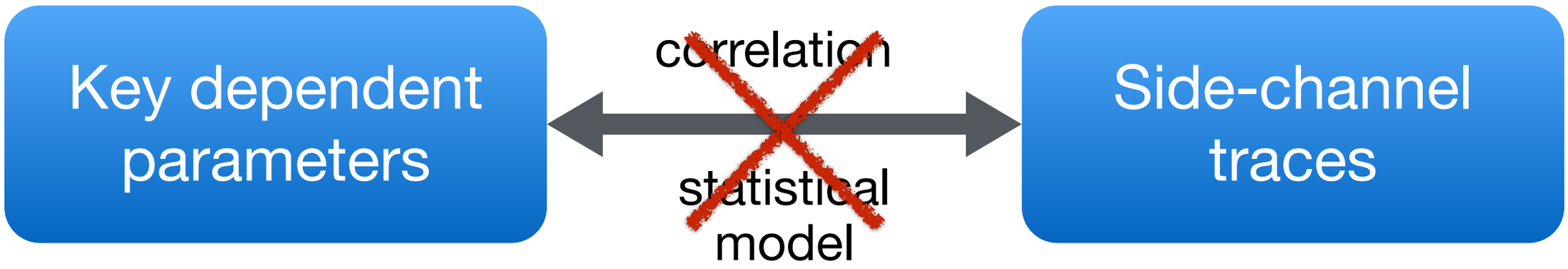
Attacks Using Side Channel Information

- **Simple — with limited number of traces**
  - Precise understanding of the implementation
  - Sufficiently high SNR
- **Differential — using statistical techniques**
  - Hypothetical modeling
  - Large amount of aligned traces
  - Correlation computation

Countermeasures

- Hiding
- Masking / Blinding
- Design partition
- Physical security

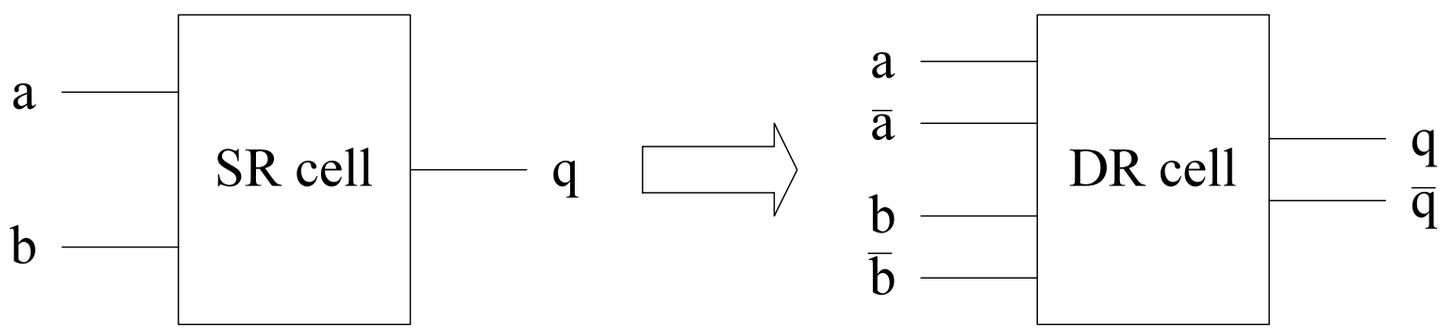
Hiding



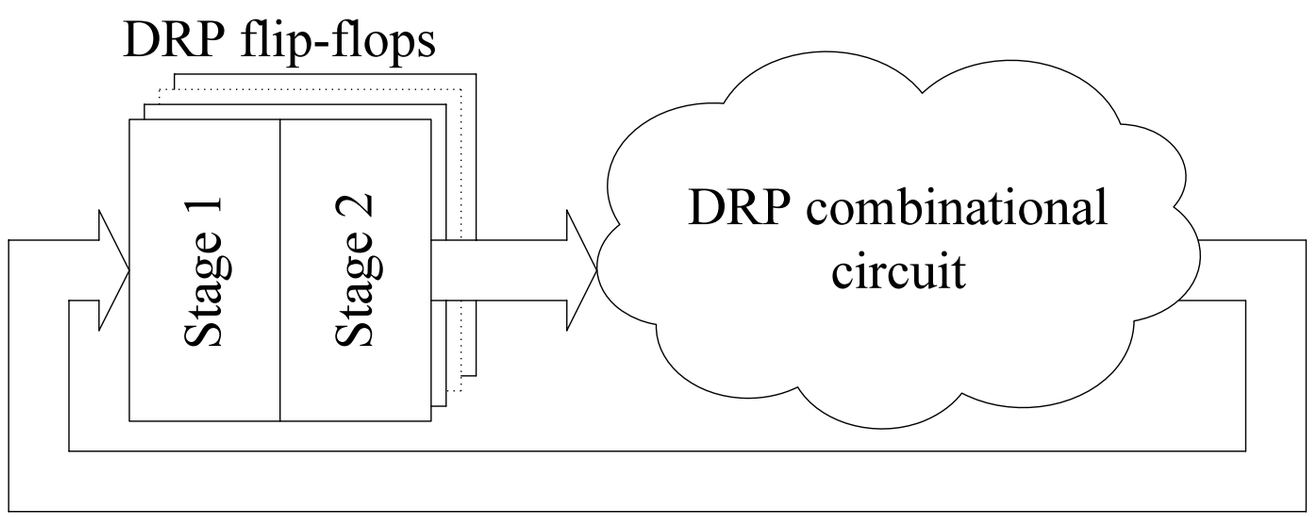
- **Lowering the SNR**
- **Noise generator**
  - Random delay
  - Circuits drawing random amount of power
  - EM noise
  - Require extra power
  - May still be susceptible to differential attacks

Balanced Logic Styles

- Dual-rail precharged (DRP) logic
  - Dual-rail logic



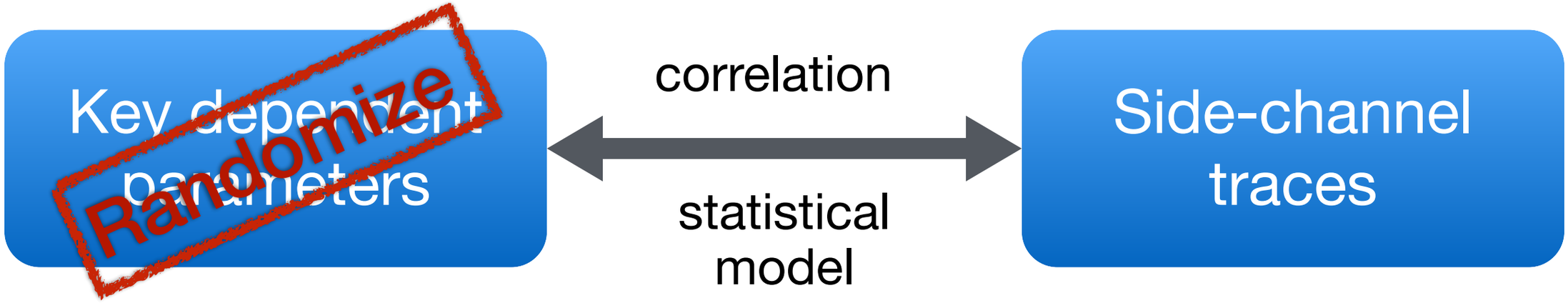
  - Adding precharge flip-flops



Other Hiding Techniques

- Low power design
- Shielding
  - Decoupling capacitors
  - EM emission shielding
  - Acoustic shielding
  - Optical shielding

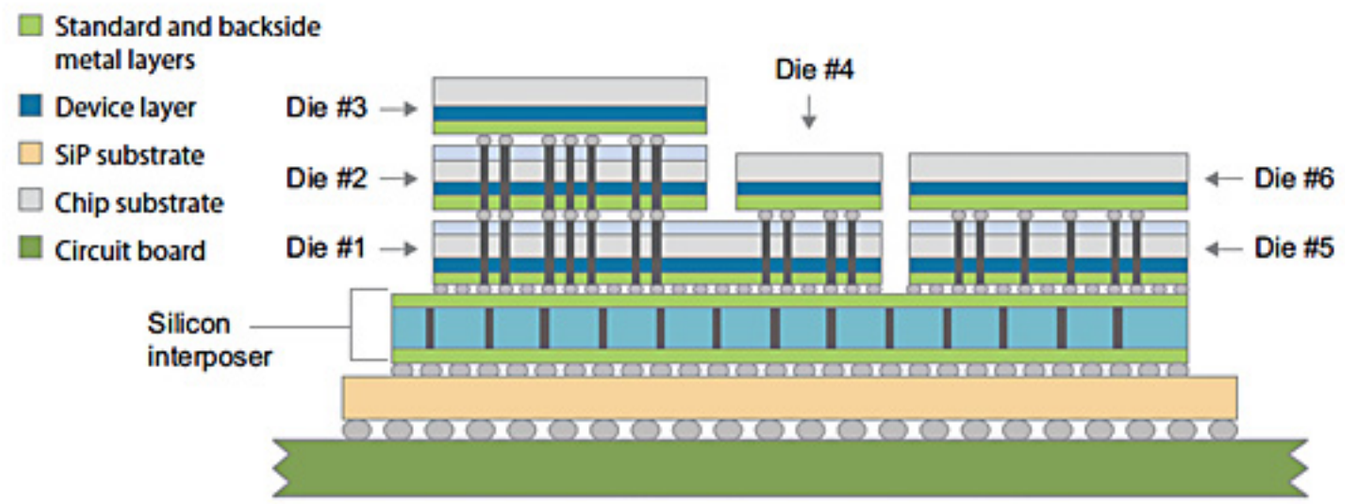
Masking / Blinding



- Randomizing the intermediate values while keeping correct crypto output
- Every intermediate value should be mixed with uniformly random masks all the time.
- Provably secure, but susceptible to high order attacks

Design Partitioning

- Separating RED (plaintext) from BLACK (ciphertext)
  - Power supply infrastructure
  - Clocking infrastructure
  - Testing infrastructure
- 3D die stacking

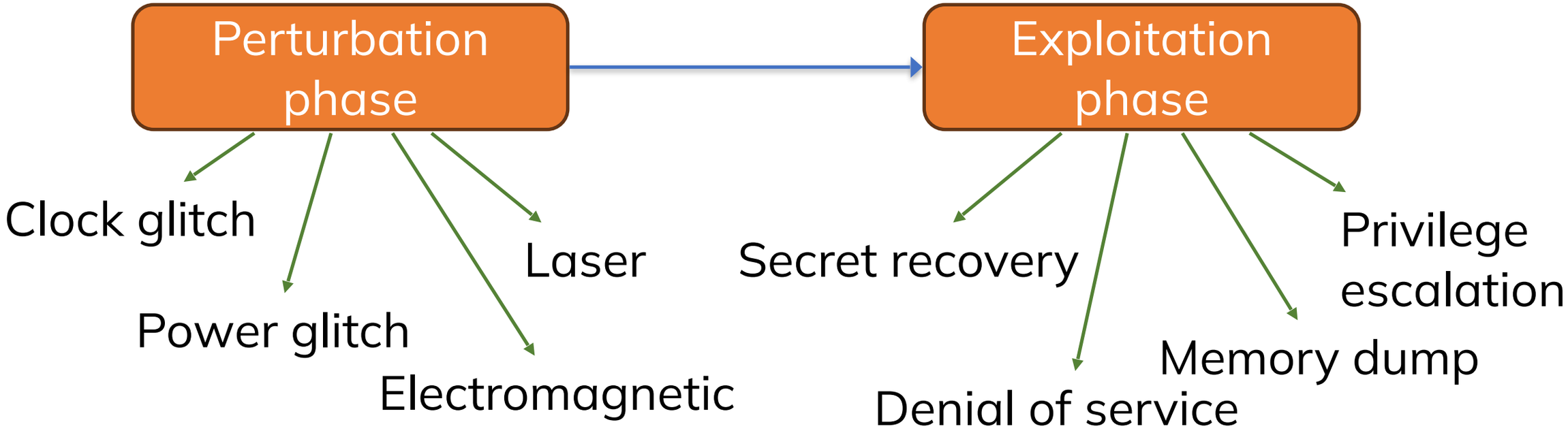


Physical Security and Anti-Tamper

- Denial of proximity, access, and possession
- Techniques to prevent / detect decapsulation

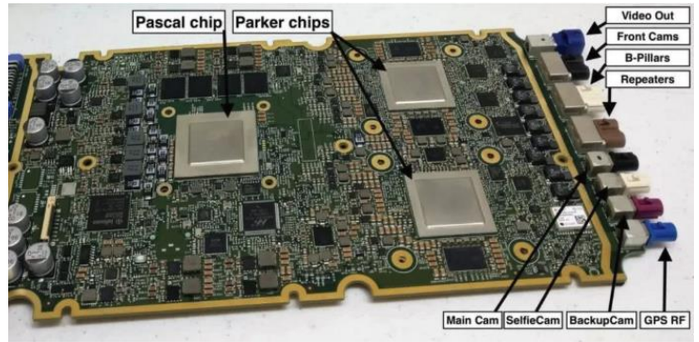
Fault Injection Attacks

- Deliberately introducing faults into system operation to create exploitable errors or bypass security guards.



Breaking the Secure Boot of Tegra X2

- An SoC designed by NVIDIA and released in 2017
- Used in devices such as
  - Magic Leap One - an XR headset
  - Tesla’s Autopilot (Rev. 2 and 2.5)
  - Skydio X2 - a military drone



- Amber Katze presented a full compromise of the Tegra X2’s secure boot at 39C3 2025.
- Combining a couple vulnerabilities in the bootloader and voltage glitching, she extracted the BootROM and found a serious flaw in the USB recovery code.

If we have enough time…

Post Quantum Cryptography (PQC)

- Public key crypto systems (PKCS) widely used today (e.g. RSA, ECC) are vulnerable to quantum attacks.
- We will need new standards for PKCS that can not be broken by quantum computers.
- PQC schemes should be able to run on today's devices.
- NIST has published the FIPS-203, 204, 205 standards for PQC on Aug 13, 2024.
- PQC ≠ quantum cryptography

Course Information

- Lecturer: Wei-Chih Hong (wchong@o365.fcu.edu.tw)
- Office: R209 IECS Building
- Lectures: 3 hours/week
- Textbook: “Power Analysis Attacks”, Mangard, Oswald, and Popp, Springer, 2007

Grading (tentative)

- **Programming assignments 40%**
  - ChipWhisperer Nano
  - Trace capture and statistical analysis
  - CPA implementation
  - Profiled attacks
- **Presentation 30%**
  - TCHES papers
- **Final CTF 30%**